



Microgesture + Grasp: A journey from human capabilities to interaction with microgestures

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ABSTRACT

Microgestures, *i.e.* fast and subtle finger movements, have shown a high potential for ubiquitous interaction. However, work to-date either focuses on grasp contexts (holding an object) or on the free-hand context (no held object). These two contexts influence the microgestures feasibility. Researchers have created sets of microgesture feasible across the entire taxonomy of everyday grasps, called transferable microgestures. However, those sets include a limited number of microgestures as compared to those for the free-hand context, for which microgestures are distinguished according to fine characteristics such as the part of the finger being touched or the number of fingers used. We provide knowledge and methods for identifying and recognizing microgestures that can transfer across contexts. First, we report a study on ergonomics factors that influence the feasibility of a microgesture in a given context. Then, we propose a conceptual model serving as a tool to determine the feasibility of a microgesture in a given context without the need for time-consuming user studies. As expected, not all microgestures were transferable to all considered contexts. Thus, we then expose two different ways of defining a set of microgestures transferable between free-hand and grasping contexts. Finally, we report a user study on recognition factors of a transferable microgesture set.

1. Introduction

Microgestures are finger based gestures defined as fast and subtle interactions (Sharma et al., 2019) that last between one (Matthies et al., 2019) and four (Ashbrook, 2010) seconds. Prior works have shown their high potential for ubiquitous interaction, *e.g.* while driving (Endres et al., 2011), cooking (Beruscha et al., 2020) or climbing (Loclair et al., 2010). Thus, microgestures can be used in a variety of situations in which our hand needs to hold objects of various shapes. The hand physical posture when holding an object is called a grasp. By their very nature, grasps constrain the number of feasible microgestures (Sharma et al., 2019; Joshi et al., 2023). Therefore, not all microgestures are transferable across all grasps. Most works on microgestures in grasp contexts focus on specific objects, thus specific grasps (*e.g.* holding a steering wheel (Wolf et al., 2011)). However, this type of focused studies makes it difficult to create a set of microgestures that can be used throughout the day, *i.e.* across a wide variety of grasps, and thus helping the wide adoption of microgestures.

A microgesture is further characterized by its action (physical motion such as a *tap* or a *swipe*), and often by its actuator (the finger performing the action), the actuator surface (the specific finger part performing the action such as the base phalanx), and the receiver (the surface touched during the action, which is very often a part of the

hand) (Soliman et al., 2018; Chaffangeon Caillet et al., 2023). While in a free-hand context, these four features have been explored and studied in details, in grasp contexts, studies focus mainly on the action (Sharma et al., 2019, 2021, 2023; Saponas et al., 2009; Wolf et al., 2011; Tan et al., 2017; Beruscha et al., 2020), the actuator (Sharma et al., 2021, 2023; Saponas et al., 2009; Wolf et al., 2011) and if the microgesture is on the hand or the object (Wolf et al., 2011; Tan et al., 2017; Beruscha et al., 2020). However, to the best of our knowledge, no study in grasp context studied the actuator surface, *i.e.* phalanxes.

In this paper, our goal is to be able to identify and recognize – with a recognition system – a set of microgestures that are transferable across all grasps. In the first part, we identify and model the ergonomics factors that impact the feasibility of a microgesture. To do so, we present a feasibility study in which we systematically tested 21 microgestures (covering 5 actions – tap, press, swipe forward, swipe inward, extend – and 3 actuators – thumb, index, middle – as well as 2 actuator surfaces for the actions tap and press – tip and middle or base –, as shown in Fig. 1S2) in 13 contexts (free-hand and the 6 grasps of Schlesinger's taxonomy considering 2 sizes of objects, as shown in Fig. 1S1) with 20 participants.

Then, we present a modelization of these results. We exposed a set of three rules, based on an analysis of the grasps characteristics, to

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determine if a microgesture is feasible based on ergonomics factors. With the grasps precisely characterized, these rules correctly predict the feasibility of microgestures 99% of the time. Our heuristic had errors only for the Lateral contexts. Due to the characteristics of the Lateral context, half of the participants contradict the theoretical analysis for 4 microgestures. In the end, we identified that 20% of the combinations context $\times$ microgesture are physically impossible or too challenging to perform, resulting in a microgesture set that is not transferable across all contexts.

In the second part, we explore how to use our results to create a recognizable set of transferable microgestures, *i.e.* microgestures that can be used in different contexts. We begin by exploring approaches for creating the most expressive set of transferable microgestures in all our contexts. To do this, the grasp set is fixed and we identify two approaches, inspired by the literature, to trade off between microgesture expressiveness and microgesture transferability. While Sharma et al. and Joshi et al. approach was to transfer the microgesture from one actuator to another to achieve complete transferability (Sharma et al., 2019; Joshi et al., 2023), we draw inspiration from Saponas et al.'s application instance to associate commands to fingers (Saponas et al., 2009) across different grasps, and merged *tap* and *press* only focusing on actuators and actuator surfaces. With this innovative approach, we identify a set of 3 microgestures transferable across all contexts. We also include 2 additional almost transferable microgestures.

Then, we build a DIY glove to recognize this set of 5 microgestures. After confirming the low impact of the glove on the microgesture feasibility with a second user study, we performed a third user study where we recognized this set 94% of the time in the 13 contexts used for the first and second experiment.

Lastly, we discuss all results from our studies at the end of this paper.

To summarize, our contributions on microgesture and grasp cover a spectrum ranging from human capabilities to microgesture interaction:

Ergonomics study: the results of the first systematic study on the feasibility of microgestures in a free-hand context and in 12 grasp contexts, covering the entire taxonomy of Schlesinger, with and without data glove on the users' hand.

Conceptual model of feasibility: the first conceptual model serving as a tool to determine if a microgesture is feasible based on a set of rules. The conceptual model relies on the concept of virtual finger and a notation for a systematic analysis.

Approaches for the creation of microgesture sets that are transferable across contexts: two documented creations of *tap*- and *press*-based microgesture sets derived from our feasibility study.

Recognition study: a recognition evaluation with an inexpensive DIY glove for a set created by considering only the actuator and actuator surface.

2. Related work

Our work relies on taxonomies of grasps for interaction and builds upon two previous microgesture research bodies that are discussed in turn in this section. The first covers microgesture-based interaction while grasping an object. The second is the body of work that investigates free-hand microgestures.

2.1. Grasp taxonomies

Grasps have been studied in many research fields, from medicine to robotics. This leads to several taxonomies of grasps. An extensive review of taxonomies prior to 1984 was published by Mackenzie and Iberall (Mackenzie and Iberall, 1994). The authors identified the 1919 Schlesinger's taxonomy (Schlesinger, 1919), which became the basis for HCI work on grasps, whether using a subset (Zhou et al., 2020;

Saponas et al., 2009) or the entire taxonomy (Sharma et al., 2019; Heumer et al., 2008; Fan et al., 2018). Schlesinger's initial goal was to group the human grasps to build functional and efficient prosthetic hands. The author focuses on the finger and palm positions and defines six grasps, as listed in Mackenzie and Iberall (1994):

Cylindrical: holding a cylindrical object, *e.g.* a mug or a hammer

Spherical: holding a spherical object, *e.g.* a pebble, a baseball ball

Palmar: holding an object with the palm facing it, *e.g.* a matchbox

Tip: holding a small object, *e.g.* a needle, a pen

Lateral: holding a flat object, *e.g.* a credit card

Hook: holding a heavy object, *e.g.* a bag

A more recent taxonomy, GRASP (Feix et al., 2016), merges 22 taxonomies of grasps, including Schlesinger's taxonomy, and is based on the following features: the force and precision to execute the grasp, how the force required to hold the object is distributed, and which part of the hand applies the force. In total, GRASP differentiates 33 grasps. GRASP includes three object-specific grasps, namely a scissor grasp, a chopsticks grasp and a cigarette grasp, that are not present in Schlesinger's taxonomy. Moreover one type of grasp in Schlesinger's taxonomy is broken down into several types in the GRASP taxonomy. To do so, GRASP identifies multiple variations of a grasp based on the number of fingers on an object, *e.g.* grasps n° 13-14-27 corresponding to the *Spherical* grasp. Further variations of a grasp rely on the size of the grasped object, *e.g.* grasps n° 1-2-3 corresponding to the *Cylindrical* grasp. To reduce the very large design space defined by the grasps, and considering both taxonomies (Schlesinger, 1919; Feix et al., 2016), we chose to use Schlesinger's taxonomy further classified into two sizes (the size being one property in the GRASP taxonomy) as was done by Sharma et al. in their elicitation study (Sharma et al., 2019) and recognition studies (Sharma et al., 2021, 2023) for microgestures in grasp contexts.

2.2. Microgesture-based interaction in a grasp context

Inspired by the design space of SoloInput (Soliman et al., 2018), we consider microgestures to be made of an action, an actuator and a touched surface.

Microgestures have already been studied in contexts where the user holds an object with the hand that performs the microgesture. Two opposing approaches have been investigated: finding specific microgestures for a given grasp (*specificity*) as opposed to finding similar microgestures across different grasps (*transferability*).

With a focus on *specificity*, previous works have created microgesture sets specific to one object such as a bike handle (Tan et al., 2017), a steering wheel (Endres et al., 2011) or a cooking pan (Beruscha et al., 2020). Wolf et al. defined a taxonomy of microgestures, with the help of biomechanics experts and physiotherapists, for three contexts of use: a steering wheel, a credit card and a pen (Wolf et al., 2011). Amongst the 21 identified microgestures, many are specific to a single object (*e.g.* a *thumb up*, only feasible with the steering wheel). However, such findings can hardly be extended to a large set of grasp contexts including the free-hand context.

With a focus on *transferability*, Sharma et al. conducted an elicitation study to associate microgestures with ten commands (*i.e.* select, delete, accept/reject, next/previous, increase/decrease, move, rotate), in a grasp context. They compared twelve objects corresponding to the six grasps of Schlesinger's taxonomy applied to two sizes of objects (Sharma et al., 2019). They identified 7 actions, *e.g.* *tap*, *press*, *swipe*, *stretch*, each triggering the same command. For instance, selection, regardless of the grasp, is performed through finger *tap*. However, for a given action, the actuator is different depending of the grasp. For instance, a *tap* is performed with the index while holding a pen (*Tip*) and with the thumb while holding a hammer (*Cylindrical*).

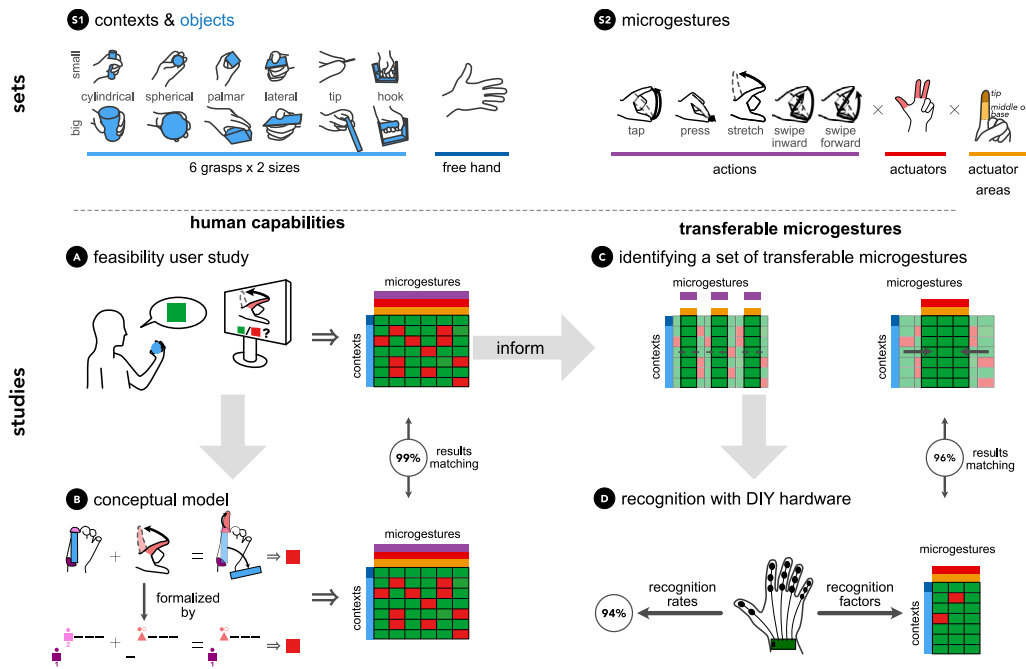


Fig. 1. S1: Sets of contexts studied in all of our studies. S2: Initial sets of microgestures studied. A: User study on the feasibility of each combination context $\times$ microgesture from S1 and S2. B: Conceptualization of the results from A. C: Designing a set of microgestures, from S2, transferable across all contexts of S1, i.e. each microgesture is feasible in each context. D: Creating a low-cost DIY glove with low impact on microgesture feasibility and high recognition rate.

In a second study, Sharma et al. address the issue of distinguishing between microgestures and noisy finger movements when interacting with everyday objects. Their first microgesture set comprises 3 actions (*tap* and *vertical swipe* in both directions) $\times$ 5 actuators that were tested for 36 scenarios with 22 daily objects (Sharma et al., 2021). After studying their feasibility, they removed the ring and the pinky finger from the possible actuators, and added 4 actions (*swipe left*, *swipe right*, *zigzag* and *circle*). They successfully recognized the final microgesture set across 5 grasps. While in theory each of the 21 microgestures (i.e. 7 actions $\times$ 3 fingers) is available for each grasp, in practice all are not feasible. However, since their focus is not on human factors, the authors do not report those impossible pairs. In our study, we want to study the feasibility of the actions presented in the elicitation study, i.e. *tap*, *press*, *stretch*, *swipes* and *draw*, for each finger and phalanx of the hand. Our goal is to identify the impossible pairs of context $\times$ microgesture.

Finally, in our daily activities, our hand either holds different objects, or not, i.e. the free-hand context. With a focus on *transferability* extended to the free-hand context, Joshi et al. propose a design space for transferable microgestures using the middle, ring or little finger as an actuator (Joshi et al., 2023). The design space is divided into two levels: hand constraints and microgesture transferability. Hand constraints are described in terms of whether the hand should be in a specific location, or not, and in a specific posture, or not. Four types of transferability are defined: across locations, across hand postures, across grasps and across fingers. In their study, they examine four actions (*tap*, *swipe* in two directions and *extension*) on two types of surface (side of finger and palm). Sharma et al. propose SparseIMU, a software to easily design a minimal IMU layout to recognize 6 actions (*tap*, *swipe* in four directions, *circle*) with 3 actuators in both free-hand and grasp contexts (Sharma et al., 2023). However, this set is not fully transferable across all their grasp contexts. Saponas et al. focus on using EMG to detect microgestures in three contexts: free-hand, holding a mug (*Cylindrical*) and holding a bag (*Hook*). They consider four actuators, index, middle, ring and pinky fingers, and a specific action for each context, *tap* in free-hand, *press* while holding a mug and *pull up* while holding a bag (Saponas et al., 2009). To demonstrate their hardware, they implemented a music application where the controls are

associated to the actuator and not the action. Depending on the context, users will perform a *tap*, a *press* or a *pull-up* with their ring finger to scroll down through the music list.

To summarize, two main approaches have been used to create transferable microgestures. With Sharma et al. (2019) and Joshi et al. (2023)'s approaches, transferability is achieved through actions, i.e. a microgesture is defined by its action and any finger can be the actuator. Moreover, since Joshi et al. (2023) only consider fingers that are not required to hold an object, namely the middle, ring and little fingers, microgestures are almost always identically transferable without changing the actuator. With Saponas et al.'s approach (Saponas et al., 2009), transferability is achieved through actuators, i.e. a microgesture is defined by its actuator and any action can be performed. Saponas et al. are the only ones in the literature to expose this approach with microgestures. However, their work does not cover the full taxonomy of Schlesinger and considers a limited set of action types.

Moreover, while the grasping microgesture literature considers only the action and actuator of a microgesture, the free-hand microgesture literature describes and studies other characteristics of a microgesture such as the phalanx touched during the microgesture or the number of fingers involved.

2.3. Microgesture-based interaction in the free-hand context

Free-hand microgesture interaction has been the subject of numerous studies in various domains: for command selection in AR (Facebook, 2021; Whitmire et al., 2017; Way, 2014; Lee et al., 2020), cockpit (Wambecke et al., 2021), rock climbing (Loclair et al., 2010), smartwatches (Chan et al., 2015; Huang et al., 2016) or for text entry in AR and smartwatches (Xu et al., 2019). M[eye]cro uses a subset of thumb-to-finger microgestures, i.e. a microgesture performed by the thumb on the other fingers, to interact with a menu in a cockpit (Wambecke et al., 2021). The authors found that microgestures have a lower impact on fatigue and tend to be faster than physical controllers. Lee et al. proposed a set of microgestures doable on finger nails for interaction with smart glasses or watches (Lee et al., 2020). They found a set of 29 usable microgestures that can be used to control widgets in augmented reality.

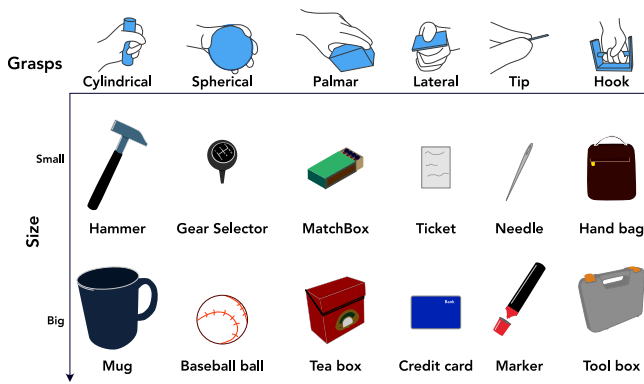


Fig. 2. List of daily objects used as reference for our 3D-printed objects.

While the above studies are based on sets of microgestures that are defined by designers, elicitation studies to define user-defined free-hand microgesture sets have been conducted and led to other types of microgestures enriching the space of possibilities. For instance, Chan et al. present a consensus on a microgesture set between 16 participants: the set includes microgestures involving unexplored characteristics such as *multi-finger taps* (Chan et al., 2016).

Building on these studies, Soliman et al. present a taxonomy of microgestures performed by a finger onto another called thumb-and-finger microgestures. The taxonomy is based on the following axes: the finger that initiates the action, i.e. the actuator, the location/surface (e.g. finger sides and phalanxes) on which the action is performed i.e. touched surface, the type of actions (e.g. *tap*, *swipe*), and the extension state of modifier fingers (Soliman et al., 2018). For instance, using Soliman et al.'s taxonomy one could differentiate a *tap of the thumb on the index base phalanx* (metacarpal phalanx) and a *tap of the thumb on the middle tip phalanx* (distal phalanx).

In summary, the literature on free-hand microgestures explores richer microgesture sets than the literature on grasping microgestures. This richness is achieved by considering finer characteristics of microgestures such as the side or the phalanx of a finger. While Joshi et al. (2023) consider the side of the finger, we want to explore the phalanx of the finger which is a commonly used characteristics in free-hand context (Wambecke et al., 2021; Soliman et al., 2018; Whitmire et al., 2017; Way and Paradiso, 2014; Way, 2014; Huang et al., 2016; Liang et al., 2021) but has never, to our knowledge, been studied in grasp context. This paper explores how to identify and recognize a set of microgestures with different actions, actuators and phalanxes that are transferable across grasp contexts covering a comprehensive set of grasps. First we study the ergonomics factors that impact the feasibility of a microgesture in free-hand and grasp contexts. Next, we define a set of transferable microgestures and focus on recognition of this set.

Part I: Study on human capabilities

3. Scope of the study: Defining contexts and microgestures

In this section, we expose the chosen contexts and microgestures.

3.1. Studied grasp contexts

As stated in the related work, we combine the taxonomy of Schlesinger (Schlesinger, 1919) and the detailed GRASP taxonomy (Feix et al., 2016). Thus, in a similar way to previous studies (Heumer et al., 2008; Sharma et al., 2019, 2021, 2023), we consider Schlesinger's taxonomy with two different sizes of objects, the size being a property in the GRASP taxonomy. For instance, for the *Tip* grasp, the two sizes make it possible to differentiate cases where one holds a needle (grasp n° 9 (Feix et al., 2016)) or a pen (grasp n° 8 (Feix et al., 2016)). In order

to test those six grasps and two sizes, we identified twelve daily objects that entice the hand to use each grasp $\times$ size possibility. Those objects are listed in Fig. 2. Sharma et al. previously reported that the object weight has little influence on the microgestures (Sharma et al., 2019), as long as it allows for comfortable grasping of the object. However, to avoid any influence from the other object's properties, such as its roughness or stiffness, we 3D printed abstract versions of these 12 objects using a Form3 resin printer.¹ The STL files of those objects are available on Github.² Fig. 1S1 shows the six grasps, two sizes and corresponding objects we consider. In the following, we call *context* a pair grasp $\times$ size.

In addition to these grasp contexts, we also consider the *free-hand* context, i.e. the hand does not hold an object. In total we consider (6 grasps $\times$ 2 sizes) + *free-hand* = 13 contexts.

3.2. Studied microgestures

First, our set includes the most common actions:

- *tap* (Whitmire et al., 2017; Song et al., 2011; Wolf et al., 2011; Saponas et al., 2009; Liu et al., 2018; Xu et al., 2019; Sharma et al., 2019, 2021; Nguyen et al., 2019; Huang et al., 2016; Kubo et al., 2019; Way, 2014; Chan et al., 2016; Soliman et al., 2018; Lee et al., 2020; Chan et al., 2015; Loclair et al., 2010; Tan et al., 2017; Wambecke et al., 2021; Liang et al., 2021): a finger comes into contact with a surface for a short period of time (similar to a mouse-click);
- *inward swipe* (Whitmire et al., 2017; Zhou et al., 2020; Beruscha et al., 2020; Song et al., 2011; Wolf et al., 2011; Liu et al., 2018; Xu et al., 2019; Sharma et al., 2019, 2021; Nguyen et al., 2019; Chan et al., 2016; Soliman et al., 2018; Lee et al., 2020; Endres et al., 2011; Loclair et al., 2010; Liang et al., 2021; Zhang et al., 2017): a finger is dragged on a surface, toward the palm;
- *forward swipe* (Whitmire et al., 2017; Zhou et al., 2020; Beruscha et al., 2020; Song et al., 2011; Wolf et al., 2011; Liu et al., 2018; Sharma et al., 2019, 2021; Nguyen et al., 2019; Chan et al., 2016; Soliman et al., 2018; Lee et al., 2020; Endres et al., 2011; Loclair et al., 2010; Liang et al., 2021; Zhang et al., 2017): a finger is dragged on a surface, away from the palm;
- *stretch* (Wolf et al., 2011; Sharma et al., 2019; Kubo et al., 2019; Way, 2014): a finger extends completely for a short period of time.

We also consider *finger press*, i.e. applying strong pressure with a finger on a surface for a short period of time. This action is promising in grasp contexts since it can be performed without moving the actuator on or off the object. Compared to the set of Sharma et al. (2019), we do not retain the *horizontal swipe* since they are mainly used with one type of actuator, the thumb (Sharma et al., 2019, 2021; Song et al., 2011; Zhou et al., 2020; Wolf et al., 2011; Nguyen et al., 2019; Chan et al., 2016; Soliman et al., 2018; Lee et al., 2020; Liang et al., 2021; Zhang et al., 2017).

Thus, this action is less generic than the others. According to Chan et al. (2016), *draw* microgestures are mainly used when no other microgesture is available. To reduce the number of microgestures, and therefore the duration of the experiment, we are focusing on the following actions: *tap*, *press*, *stretch* and *inward/forward swipe*.

Beyond the performed actions, we consider two additional microgesture characteristics to enrich the set of microgestures.

First, we distinguish the actuator of the microgesture. Wolf et al. (2011) and Sharma et al. (2021) reported difficulties for the *pinky* and *ring* fingers to perform microgestures on an object due to biomechanical

¹ <https://formlabs.com/fr/3d-printers/form-3>

² <https://github.com/AChaffangeon/schlesinger-abstract-objects.git>

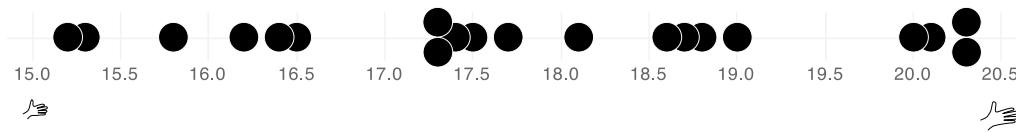


Fig. 3. Hand size distribution (distance in cm between the base of the palm and the middle tip).

restrictions. So we decided to remove the two fingers from our set of microgestures.

Second, for *tap* and *press*, we distinguish the area of the actuator that taps/presses the surface: the *tip phalanx* and the *middle or base phalanxes* (mob). We decided to merge the middle and base phalanxes because, in a pilot study, we noticed that both phalanxes worked together. A *tap* of the middle phalanx on a large object almost always results in a *tap* of the base phalanx, e.g. *Cylindrical Big* (Fig. 2). A *tap* of the base phalanx in some contexts, e.g. *Palmar Big* or *Hand Free* (Fig. 2), is uncomfortable.

To summarize, the set is composed of:

(tap, press) $\times$ (3 fingers $\times$ 2 phalanxes) + (stretch, inward, forward) $\times$ 3 fingers = 21 microgestures.

4. Feasibility: Ergonomics factors

We present an experimental study to determine the feasibility and ease of use of the above set of microgestures when performed in the thirteen defined contexts.

4.1. Design

We designed a 13×21 within-subjects experiment with the independent variables *CONTEXT* and *MICROGESTURE*. We randomized the order of *CONTEXT*. For each *CONTEXT*, we randomized the order of *MICROGESTURE*. Each participant tested each pair of *CONTEXT* $\times$ *MICROGESTURE* once.

The dependent variables are: if a pair *CONTEXT* $\times$ *MICROGESTURE* is SKIPPED or not and the *EASINESS* of the pair. We also recorded the type of *SURFACE* on which the microgesture was performed.

4.2. Measurements

Participants could skip a trial, using a keyboard key, if the pair *CONTEXT* $\times$ *MICROGESTURE* was not feasible. If the trial was not skipped, then we recorded the *EASINESS* of the pair using a 4-point Likert Scale (1 - hard, 4 - easy). We used an even number of graduations to avoid neutral answers. Finally, for the *SURFACE*, participants selected one of the three surface types from Sharma et al. (2019), namely *body*, *object* and *air*. Given our microgesture set, only *stretches* can be performed in the air, i.e. *SURFACE* = *air*.

4.3. Task

After presenting the purpose of the study, the participants sat down in front of a computer. For each *CONTEXT*, the experimenter explained to the participants how to hold the object. The experimenter then told the participants that they could perform the microgestures as they wished, but that they could not modify their grasp. For each pair *CONTEXT* $\times$ *MICROGESTURE*, a picture depicting the *CONTEXT* and a picture depicting the *MICROGESTURE* were displayed on the screen. Participants could then try to perform the microgesture. Then, they could use the keyboard either to skip the microgesture or to enter the easiness of the microgesture and the surface on which the microgesture was performed. After skipping the microgesture or entering the easiness and surface of the microgesture, the next pair *CONTEXT* $\times$ *MICROGESTURE* was automatically displayed.

In all, the experiment lasted less than an hour.

4.4. Participants

We recruited 20 participants (mean age = 26.4 yo, $\sigma = 6.0$), all using their right hand for daily interaction. Ten participants were biologically females and ten were males. We ensure to cover a broad range of hand sizes (15.3 cm to 20.3 cm between the base of the palm and the middle tip). Fig. 3 shows the hand size distribution.

4.5. Results

The data analysis follows recommendations for “fair statistical communication in HCI” (Dragicevic, 2016). Data were aggregated by user using the median, then, if needed, aggregated by context and/or microgesture using the mean. For Likert score comparison, we report, as an effect size, Cliffs δ . Cliffs δ represents how often the Likert scores from a group A are greater than scores from a group B. δ ranges from -1 to 1 . A score around 0 means no effect. A score close to 1 means a large effect with Likert scores from A being always greater than those from B. A score close to -1 means a large effect with Likert scores from B being always greater than those from A. We computed Cliffs δ using the Python library `cliffs-delta`.³

Fig. 4 presents the number of participants who felt that a context $\times$ microgesture combination was impossible to perform.

Fig. 5 presents the median Likert score (scale from 1-hard to 4-easy) for each context $\times$ microgesture combination. If a participant skipped a combination, no Likert score was recorded. If 50% or more participants skipped a combination, then the combination was deemed impossible. Median scores are therefore only calculated for combinations performed by at least 50% of the participants.

4.5.1. Hand size impact

We plotted the number of skipped trials and the median score for each hand size. The percentage of skipped context $\times$ microgesture combinations appears to fit a polynomial curve of degree 2 (p -value = 0.01 , $R^2 = 0.41$), see Fig. 6. Since the extremities of the parabolic curves tend towards infinity, having a small or big hand could impact the feasibility of a microgesture. However, if we look at the confidence band, there does not seem to be any effect in our range of hand sizes. Based on comments from our participants with smaller hands, we expected an impact of the hand sizes on thumb microgestures for two contexts: *Spherical Big* and *Palmar Big*. After plotting the data, see Fig. 7, there does not seem to be any impact for *Palmar Big* because when participants with small hands cannot perform the microgestures, participants with larger hands cannot either. For the *Spherical Big*, there appears to be an influence. The *Spherical Big* object is supposed to be held between the fingers and the palm without the need of the thumb. However, four participants (P2, P8, P19, P20), with hand length under 16.5 cm, needed the thumb to hold the object properly. These participants could not perform microgestures with the thumb in this context. However, two participants (P9, P15), who also have a hand length under 16.5 cm, could perform all thumb microgestures in this context. Thus, hand size, and possibly thumb length, can impact the feasibility of microgestures depending on whether users can hold the object with or without the thumb.

Similarly, we did not notice any impact of hand size on Likert scores, see Fig. 8, since each confidence interval intersects the others by more than a quarter of their length.

³ <https://pypi.org/project/cliffs-delta/>

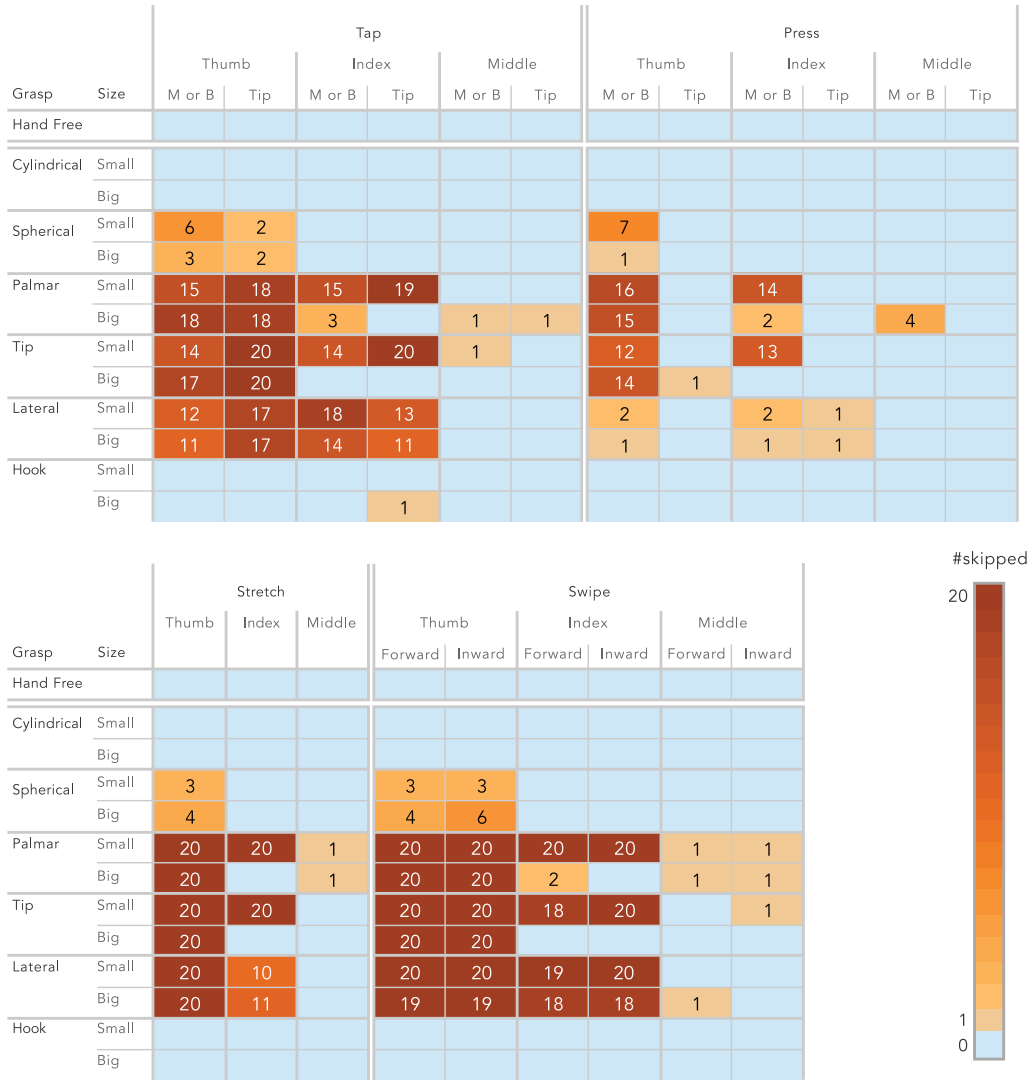


Fig. 4. Number of participants who skipped a context × microgesture combination in Study 1.

4.5.2. Surface

As a general rule, a finger that is already in contact with the object will preferably, if not exclusively, perform microgestures on the object, and otherwise on the hand. This was surprising for *Cylindrical Small* and both *Hook* contexts. In these contexts, the thumb can move freely and easily reach the other fingers. We expected participants to *tap/press* on the thumb for microgestures with the tip phalanx as we felt that the thumb is easier to reach than the object. We believe that since a majority of microgestures had to be performed on the object given that the thumb was not reachable/movable, users performed all microgestures on the object. Based on these findings, when investigating microgestures for a specific object, and when the actuators are already in contact with the object, we recommend augmenting the object rather than equipping users with recognition devices, as done in Song et al. (2011).

4.5.3. Grasp impact on microgesture

Overall, the confidence intervals suggest that grasp has an effect on the easiness of a microgesture, see Table 1.

Based on the GRASP taxonomy, we further study the impact of grasp on microgesture feasibility by considering 4 categories: gravity, palm, pad and mixed. These categories are built based on the virtual fingers, i.e. units of one or more part of the hand which apply forces in the same direction (Arbib, 1985; Iberall, 1987) to grasp an object. A grasp

is typically composed of at least two virtual fingers (VF). In the GRASP taxonomy, authors created a table describing the virtual fingers for 47 grasps of the literature (Feix et al., 2016).⁴

In gravity grasps, there is no virtual finger as the object lies on the hand. We have only tested one version of this grasp category with the object lying on four fingers: *Hook Small* and *Hook Big*. In these contexts, every microgesture is feasible with ease (4.0 ± 0 , $n = 21$). However, to be able to perform *press* with the fingers, the thumb must be on the object and opposite the finger that presses.

In palm grasps, the object is held between the palm (VF1) and two or more fingers (VF2). This is the case for *Cylindrical Small*, *Cylindrical Big* and *Spherical Big*. In these contexts, everything is feasible with ease (3.9 ± 0.2 , $n = 21$).

In pad grasps, the object is held between the thumb (VF1) and a group of 1 or more fingers (VF2). This is the case for *Palmar Small*, *Palmar Big*, *Tip Small* and *Tip Big*. Thumb microgestures are impossible to perform with the exception of *tip press*. If there is only one finger in VF2, as in the smaller grasps, microgestures with that finger are impossible with the exception of *tip press*. Otherwise, all microgestures can be performed with fingers other than the thumb. Based on confidence intervals, there seems to be no difference in the ease of

⁴ http://grasp.xief.net/documents/taxonomy_comparison.pdf

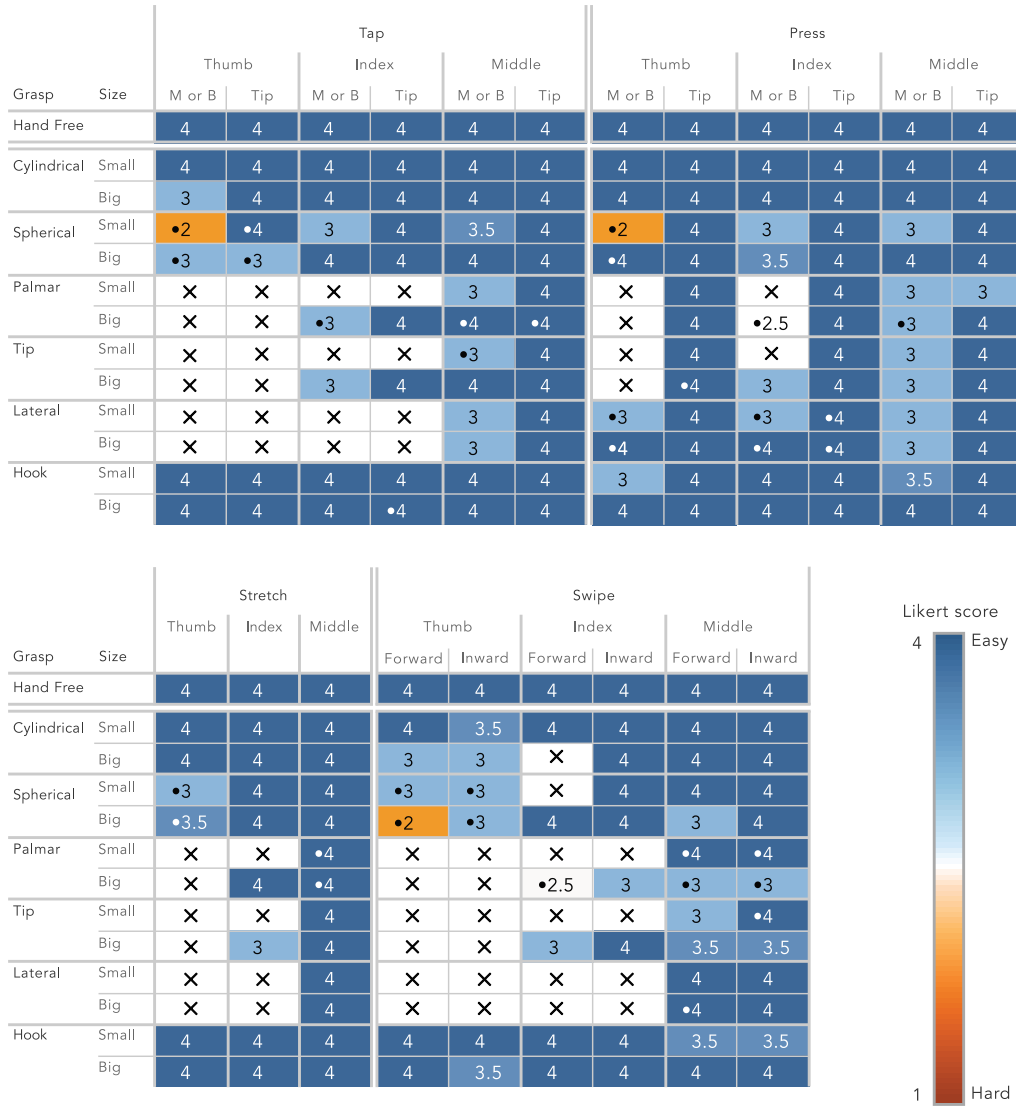


Fig. 5. Median Likert score (scale from 1-hard to 4-easy) for each context × microgesture combination in Study 1. Cells with a × symbol correspond to context × microgesture combinations that were skipped by 50% or more participants. Cells with a • symbol correspond to combinations that were skipped at least once.

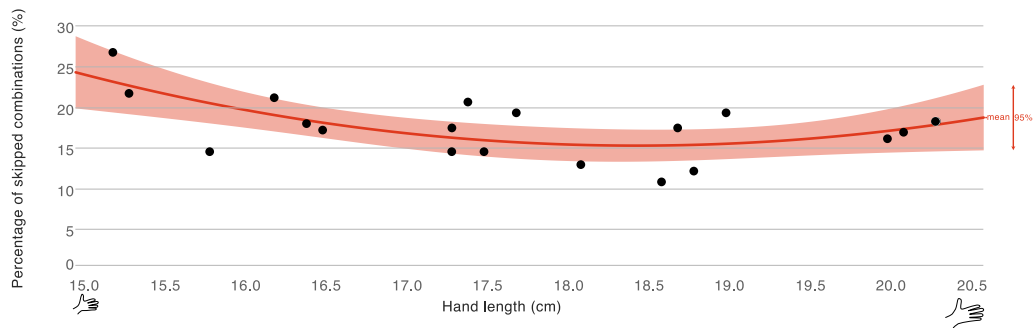


Fig. 6. Mean number of skipped context × microgesture combinations per hand size with a polynomial fit and the 95% confidence band. Each dot corresponds to a unique user, for the hand size 20.3, two dots are superimposed.

microgestures between pad grasps (3.5 ± 0.3 , $n = 18$) and palm grasps (3.9 ± 0.2 , $n = 21$).

According to the GRASP taxonomy, *Spherical Small* is a pad grasp where the thumb (VF1) is in opposition to the 4 fingers (VF2). However, thumb microgestures were feasible with the *Spherical Small* object.

The reason is that users held the object with the thumb in opposition to the four other fingers, but, when users needed to perform thumb microgestures, they spread their fingers across the object and held it between the index and pinky fingers only. As this does not conflict with Schlesinger's taxonomy definition of *Spherical* grasp, we have allowed

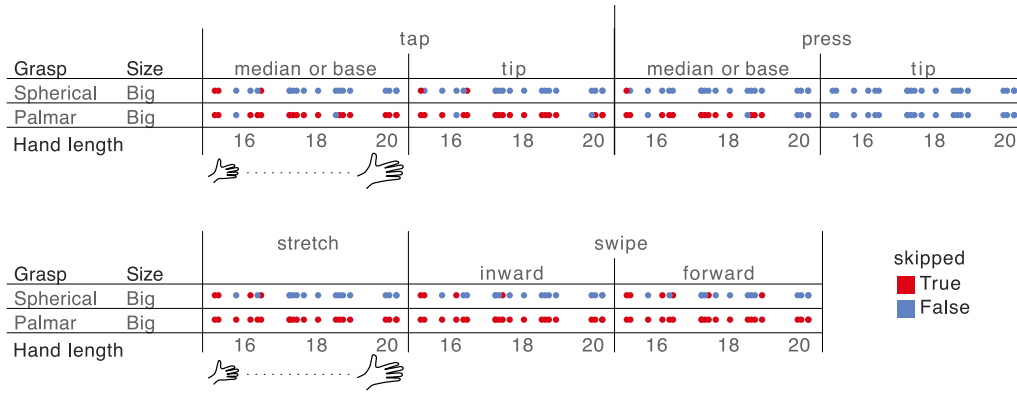


Fig. 7. Distributions of skipped (red) and non-skipped (blue) microgestures, performed with the thumb, across hand lengths for *Spherical Big* and *Palmar Big* contexts. Each dot in each *hand-length* cell corresponds to one participant. For instance, no participant was able to perform the *stretch* microgesture in *Palmar Big*, while the participants with the smallest hands were the only ones unable to perform it in *Spherical Big*.

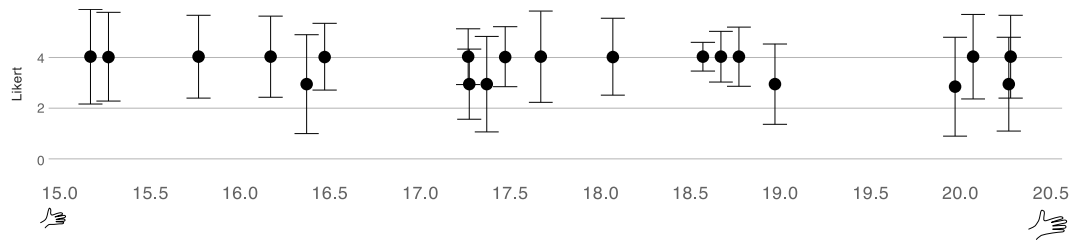


Fig. 8. Mean Likert with 95% confidence interval per hand length. Each hand length corresponds to a unique user.

Table 1

Mean median scores of the microgestures per grasp with 95% confidence interval (CI).

Grasp	Size	Mean (CI)	Grasp	Size	Mean (CI)
Hand Free		4.0 ± 0			
Cylindrical	Small	4.0 ± 0.0	Spherical	Small	3.5 ± 0.3
	Big	3.9 ± 0.2		Big	3.7 ± 0.2
Palmar	Small	3.8 ± 0.3	Tip	Small	3.7 ± 0.3
	Big	3.5 ± 0.3		Big	3.7 ± 0.3
Lateral	Small	3.6 ± 0.3	Hook	Small	3.9 ± 0.1
	Big	3.8 ± 0.3		Big	4.0 ± 0.0

users to do this method. In the end, *Spherical Small* is a pad grasp with two pairs of VF, thumb-middle&ring and index-pinky. In cases where there are several pairs of VF, all microgestures are feasible.

In the mixed grasp category, the grasp shares some properties between pad and palm. This category also includes all grasps that do not fall into the three above-mentioned categories. We have only tested one version of this grasp category, the *Lateral* grasp. Because of the great variability between grasps in this category, e.g. a grasp for holding chopstick or a grasp for holding a cigarette (Feix et al., 2016), we recommend not generalizing the results from the *Lateral* grasp to the other grasps in this category. Nonetheless, with the two *Lateral* grasps, thumb and index microgestures are not possible with the exception of *press*.

5. Feasibility: Conceptual rules-based model

All in all, determining the feasibility of microgestures in different contexts requires to setup an experiment, which can be tedious and costly in terms of time and effort. In this section, we conceptualize the results of our user study to create a set of rules for determining whether or not a microgesture is feasible in a given context, without the need for an experiment.

5.1. Rules for microgesture feasibility

Through an analysis of virtual fingers in a context, we propose a systematic approach to infer if a context × microgesture is feasible or not.

First, according to the definition of a grasp in the GRASP taxonomy, **Rule 1:** a grasp, other than gravity grasps, must have at least two VF units available at all times during a microgesture.

Second, looking at the impossible microgestures for the pad grasps, it seems that if a single finger is used as a VF it can only *press* with the tip of that finger. However, if two fingers belong to the same VF, all microgestures are possible with both fingers. From this, we derive the following two rules.

Rule 2 A VF unit must have at least one finger that does not move on or off a surface at any time during a microgesture.

Rule 3 A single finger used as a VF can only perform a *press* gesture with the phalanx that is already in contact with the object.

5.2. Describing the feasibility rules using μ glyph

To ease the use of the aforementioned rules, we represent the rules using the notation μ Glyph (Chaffangeon Caillet et al., 2023). μ Glyph is a notation to precisely define microgestures.

With μ Glyph, a microgesture is described using the movement, or absence thereof, of a finger and contact information. Using a graphical

Table 2

Visual representation of the virtual fingers for the 12 grasps. The virtual finger analysis comes from the GRASP taxonomy (Feix et al., 2016). The Hook grasp has no virtual finger since the object is not held but rests on the hand thanks to gravity (Feix et al., 2016).

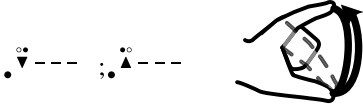
	Cylindrical	Spherical	Palmar	Lateral	Tip	Hook
Small						×
Big						×

Table 3

μ Glyph descriptions of the studied microgesture set with x being either the thumb, index or middle finger and y being either the tip or mob phalanx.

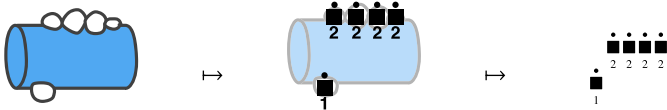
Tap	Press	Inward Swipe	Forward Swipe	Stretch
$x^{\circ}y; x^{\circ}y$	$x^{\circ}y; x^{\circ}y$ Lo $\rightarrow$ Hi Hi $\rightarrow$ Lo	$x^{\circ}y; x^{\circ}y; x^{\circ}y$	$x^{\circ}y; x^{\circ}y; x^{\circ}y$	$x^{\circ}y; x^{\circ}y$ x $\rightarrow$ Op

structure, we can specify the actuator, *i.e.* the finger performing the microgesture, and the contact surface, *i.e.* the finger or object on which the microgesture is performed. For instance, an *index tap on the thumb* can be represented as follows:

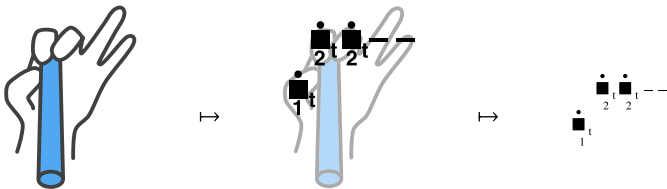


By reusing the μ Glyph notation and adding simple number indices, virtual fingers can be represented visually, with each number representing one VF unit. These VF indices can only be used for a finger that does not move on or off the object, *i.e.* a finger whose context is $\bullet$.

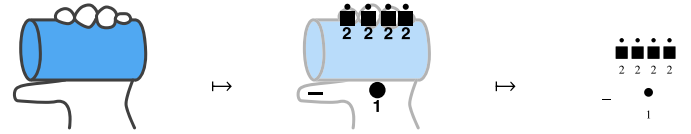
Using μ Glyph and these VF indices, we can represent a grasp where one holds a mug between the thumb and other fingers as follows:



We use the *still + contact* ($\bullet$) glyph with an x index for a finger that is part of a virtual finger unit. To be more precise, we can, according to μ Glyph, specify which particular phalanx of the finger is in contact with an object. For instance, a pen held only by the tips (t symbol) of the thumb, index and middle fingers can be represented as:



In GRASP (Feix et al., 2016), the *Cylindrical Big* object is considered to be held between the palm and the four opposite fingers. For the palm, we use the contact glyph $\bullet$ and add the VF indices under this symbol. Thus, the context *Cylindrical Big* is represented by:



Using this extended μ Glyph notation, Tables 2 and 3 summarize all studied contexts and microgestures. For each studied grasp, we used the VF analysis proposed in the GRASP taxonomy (Feix et al., 2016).⁵

Using this notation, the aforementioned rules can be written as:

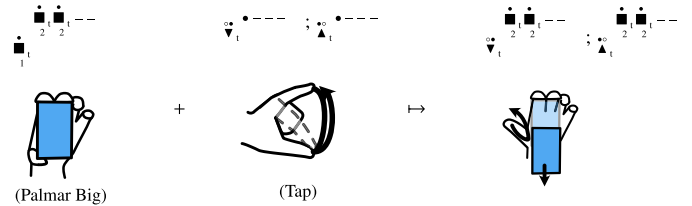
Rule 1: there must be at least two different VF indices in the μ Glyph description of the pair context $\times$ microgesture for the microgesture to be feasible.

Rule 2: a VF unit must have one finger whose μ Glyph's context (*i.e.*, the glyph exponent) is $\bullet$.

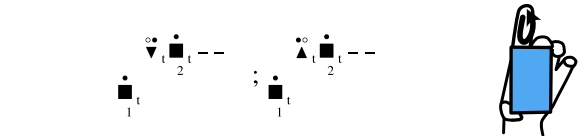
Rule 3: If a single finger is used as a VF, it can only perform microgestures that can be described by $\overline{}_x$, with x a phalanx already in contact with the object. In this case, such finger can only perform a *press* microgesture (∇ ; $\blacktriangle$).

5.3. Applying the feasibility rules

Applying these rules and combining the μ Glyph expressions of a context and a microgesture, we can infer if a combination is feasible or not. For instance, the microgesture *thumb tip tap* in the context *Palmar Big*, is represented as follows:



Since there is only one VF unit left, the object will fall. However, an *index tip tap* microgesture in the *Palmar Big* context, is represented as follow:



Since there are still two VF units left, we consider a *index tip tap* to be feasible in a *Palmar Big* context.

Fig. 9 summarized the results of applying this conceptual rules-based model to the microgestures and grasps of the feasibility study.

⁵ http://grasp.xief.net/documents/taxonomy_comparison.pdf

Grasp	Size	Tap						Press					
		Thumb		Index		Middle		Thumb		Index		Middle	
		M or B	Tip	M or B	Tip	M or B	Tip	M or B	Tip	M or B	Tip	M or B	Tip
Hand Free													
Cylindrical	Small												
	Big												
Spherical	Small	•	•					•					
	Big												
Palmar	Small												
	Big												
Tip	Small												
	Big												
Lateral	Small							•		•			
	Big							•		•			
Hook	Small												
	Big												

Grasp	Size	Stretch			Swipe					
		Thumb	Index	Middle	Thumb		Index		Middle	
					Forward	Inward	Forward	Inward	Forward	Inward
Hand Free										
Cylindrical	Small									
	Big									
Spherical	Small	•			•	•				
	Big									
Palmar	Small									
	Big									
Tip	Small									
	Big									
Lateral	Small									
	Big									
Hook	Small									
	Big									

Fig. 9. Conceptual analysis of the feasibility of microgestures for each context. Black boxes represent conceptually impossible context × microgesture combinations. The green symbols •, and blue symbols •, represent the limits of the model: incorrect VF analysis and changing VF, respectively. Given that the context of the Hook cannot be represented with virtual fingers, because the object is held in the hand by gravity, we deduce that all microgestures are feasible. Similarly, we deduce that all microgestures can be performed with the free-hand.

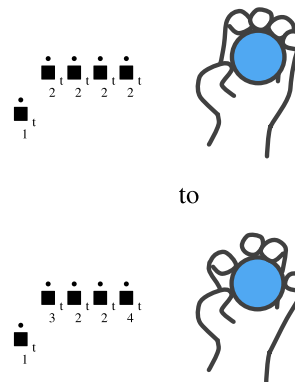
5.4. Validation of the conceptual rules-based model

To validate the conceptual model, we compare the results of applying the rules to our contexts and microgestures, *i.e.* Fig. 9, with the results of our empirical experiment, *i.e.* Fig. 4.

Out of the 273 combinations of context × microgesture predicted in the conceptual analysis, 263 were correctly determined (96%). The 11 incorrectly determined combinations correspond to combinations deemed impossible that were actually feasible. These combinations are only for three contexts: Spherical Small and both Lateral Small and Big.

5.5. Limitations of the conceptual rules-based model

Incorrect VF analysis. As explained in the results of the feasibility experiment (Section 4.5.3), users adjusted their grasp with the *Spherical Small* object to perform microgestures. In fact, users slightly moved their index and pinky fingers on the sphere to perform microgestures with the thumb, see Fig. 10A. This slight change in the way the small spherical object is held alters the VF analysis from



where VF1 is opposed to VF2 and VF3 is opposed to VF4. With this new grasp context, the thumb can lift off the object since VF3 and VF4 are sufficient to hold the object.

Nonetheless, considering this new VF analysis for the *Spherical Small* context, we correctly determined 99% of the time if a combination was feasible or not.

Changing VF. In both *Lateral* contexts, two microgestures were actually feasible: *thumb middle or base press* (mob), *index mob press*. The *thumb mob press* was done by shifting the force from the tip to the

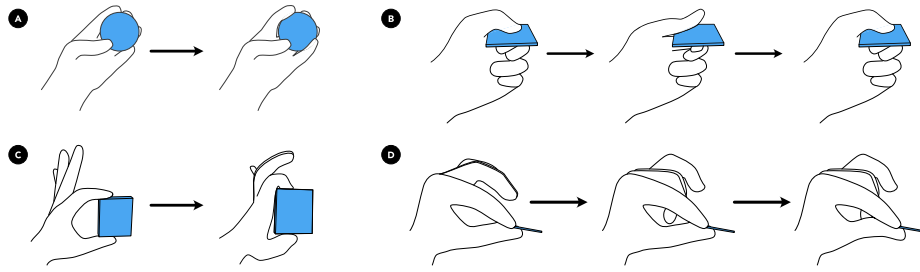


Fig. 10. (A) participants holding the *Spherical Small* object with their index and pinky finger on the sides of the object to perform *thumb* microgestures. (B) participants shifting their VF from thumb tip to thumb mob allowing them to perform *thumb mob press* in both *Lateral* contexts. (C) participants moving the object to *tap* on the *mob* phalanx of the index with the object. (D) participants moving the middle finger to *tap* on it with the thumb.

mob phalanx of the thumb, thus shifting VF1 from the thumb tip to the thumb mob, see Fig. 10B. The index mob press was done by tilting the thumb on the object to be either in front of the index tip or index mob phalanx, thus shifting VF2 from the index tip to index mob. Therefore, in both cases, participants managed to change for a brief moment their VF without dropping the object.

VF can move with the object. Our rules consider that the fingers that are part of a VF cannot move on or off the object, which implies that the fingers must be static. However, in *Palmar Small* and *Tip Small* contexts, some participants managed to do a *index mob tap* by bringing the object on the mob phalanx, see Fig. 10C. Thus, the fingers that are part of a VF are static relatively to the object but they can move the object to reach other parts of the hand. Nevertheless, this represents a minority of participants who have tried to always find a way to perform the microgestures. Therefore, the conceptual analysis still worked for these context $\times$ microgesture pairs.

Fingers that are not the actuator can move. The rules only take into account the fingers that are part of a VF and the actuator. However, in *Tip Small* and *Palmar Small*, to perform microgestures with the thumb mob phalanx, participants brought a finger close to the phalanx, e.g. the middle or ring finger, then moved the thumb while holding the object, see Fig. 10D. As the previous limitation, this represents a minority of participants.

In the end, the conceptual model seems reasonably relevant and models the average behavior observed in the empirical study, including for the context $\times$ microgesture combinations mentioned just above.

Part II: Study on the recognition of transferable microgestures

6. Scope of the study: Designing a set of transferable microgestures

As shown in Fig. 4 some microgestures in our set are not transferable across all the grasp contexts. In what follows, we explore ways of creating a large set of transferable microgestures.

We will consider *press*, as it is the only microgesture feasible with all fingers in each grasp, and *tap*, as it is the most common microgesture in the literature. As it is not possible to use all actuators (*i.e.* fingers performing microgesture) and all specific actions (*i.e.* the movement of the microgestures) for transferability, we will explore two ways, inspired by the literature: focusing on actions, and focusing on actuators.

6.1. Action-based set

Sharma et al. (2019) and Joshi et al. (2023) focused on specific actions (*tap* (Sharma et al., 2019; Joshi et al., 2023), *press* (Sharma et al., 2019), *swipe* (Sharma et al., 2019; Joshi et al., 2023), *extension* (Sharma et al., 2019; Joshi et al., 2023) and *draw* (Sharma et al., 2019)) and decided not to consider any specific actuator for complete transferability. They were thus able to define a set of microgestures transferable across different grasp contexts. Indeed, for the same grasp, each microgesture can be performed by different actuators and combination of actuators. To illustrate this approach, let us take an example from Sharma et al.

Grasp	Size	Tap		Press	
		Base	Fingertip	Base	Fingertip
Hand Free		4 m	4 m	4 m	4 m
Cylindrical	Small	4 m	4 m	4 m	4 m
	Big	4 m	4 m	4 m	4 m
Spherical	Small	3.5 m	4 m	3 m	4 m
	Big	4 m	4 m	4 m	4 m
Palmar	Small	3 m	4 m	3 m	4 m
	Big	4 m	4 m	3 m	4 m
Tip	Small	3 m	4 m	3 m	4 m
	Big	4 m	4 m	3 m	4 m
Lateral	Small	3 m	4 m	3 m	4 m
	Big	3 m	4 m	3 m	4 m
Hook	Small	4 m	4 m	3.5 m	4 m
	Big	4 m	4 m	4 m	4 m

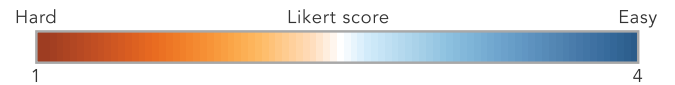


Fig. 11. Action-based set. The best actuator(s) for each microgesture is(are) represented on the side of the cell. t: thumb. i: index. m: middle.

(2019): while holding a mug (*Cylindrical Big*), a *tap* is performed by the thumb while a *press* is performed by the index, middle, ring and pinky finger altogether. Therefore, considering only *tap* and *press*, this approach results in a set of 2 microgestures.

Since our study takes into account two actuator areas, we can extend this set of transferable microgestures: 2 actions (*tap*, *press*) $\times$ 2 actuator areas (tip phalanx, mob phalanx). Since an object can be hold with two virtual fingers (Rule 1), all microgestures are transferable across all contexts because there is always at least one finger that is not alone in a virtual finger (Rules 2 & 3). Based on the results of the experimental study (Fig. 5), Fig. 11 presents the median Likert score for each context $\times$ microgesture combination, indicating the best actuator(s) for each combination. The median Likert score for this set is 3.8 ± 0.1 . Thus, by considering actions and actuator areas, we obtain a set of 4 microgestures transferable across all contexts.

6.2. Actuator-based set

Saponas et al. used a set of 4 microgestures considering 4 actuators (index, middle, ring and pinky) that could perform either a *tap* or a *press* (Saponas et al., 2009). Inspired by their method, if we disregard the action performed, but retain the distinctions between actuators and between actuator areas, we obtain the set: 3 actuators (thumb, index, middle) $\times$ 2 actuator areas (tip phalanx, mob phalanx).

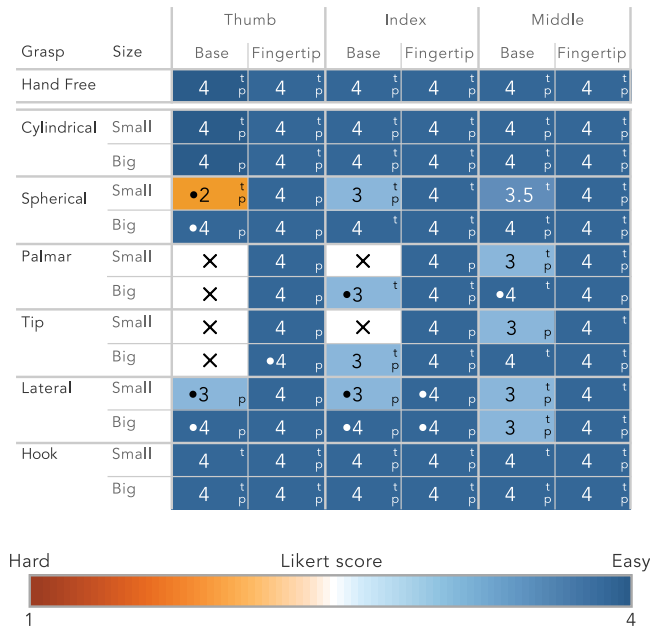


Fig. 12. Actuator-based set. The best action is represented on the side of the cell. t: tap. p: press.

In this set, the mob phalanx is not transferable to the *Palmar* and *Tip* contexts where an actuator alone in a virtual finger has only the tip phalanx in contact with the object (Rule 3). Based on the results of the experimental study (Fig. 5), Fig. 12 presents the median Likert score for each context $\times$ microgesture combination, indicating the best action(s) for each combination. The median Likert score for the whole set is 3.8 ± 0.1 . Thus, considering actuators and actuator areas, we obtain a set of 6 microgestures, 3 of which are transferable across all contexts (tip phalanx of thumb, index and middle fingers), 2 are transferable across 11/13 contexts (mob phalanx of index and middle fingers) and 1 is transferable across 9/13 contexts (mob phalanx of thumb).

By integrating the idea of combining actuators introduced in the previous method by Sharma et al. (2019), and which is anchored in the free-hand microgesture literature (Chan et al., 2016), we can expand our actuator-based set of transferable microgestures. We therefore propose to add a microgesture performed simultaneously by all actuators: a simultaneous *tap* (or *press*) of the tip phalanxes of the thumb, index and middle fingers. This gives us a final set of 7 microgestures, of which 4 are transferable across all contexts, 2 are transferable across 11/13 contexts and 1 is transferable across 9/13 contexts.

Having defined two sets of microgestures that are transferable across free-hand and grasp contexts, we now focus on recognizing such microgestures. In the following sections, we present a DIY glove for capturing microgestures. After verifying the impact of the glove on microgesture feasibility, we will present a recognition mechanism to recognize the actuator-based set.

7. Building recognition hardware for transferable microgestures

In this section, we expose our low cost and portable recognition hardware to recognize microgestures. We also present a study on the impact of wearing such a glove on the feasibility of microgestures.

7.1. Low cost and portable recognition hardware

While there are many different kinds of hardware to recognize microgestures, e.g. cameras (Way, 2014; Sharma et al., 2021; Soliman et al., 2018; Zhou et al., 2020), gloves (Wambecke et al., 2021;

Whitmire et al., 2017), rings (Zhang et al., 2017; Liang et al., 2021), exo-skeleton with IMUs (Sharma et al., 2023), electromyography arm-band Saponas et al. (2009), to our knowledge, only gloves offer the possibility to easily capture pressure information (Wambecke et al., 2021; Whitmire et al., 2017) without augmenting the objects. The prototype from M[eye]cro (Wambecke et al., 2021) uses commercial pressure sensors. However, due to the rigidity of these sensors, all closing movements were physically constrained reducing the range of motion. Moreover, the size and shape of the sensors have resulted in some parts of the fingers not being covered. The prototype from Digitouch (Whitmire et al., 2017) uses a two-part sensor: an emitter part on the thumb and a receiver part on each finger. In our case, the user will perform most microgestures on the object rather than on the thumb, as exposed in our first study, thus those sensors are unusable with non-augmented objects in the hand. We therefore opted to make our own custom sensors that work on any surface, properly covering the phalanges without imposing physical resistance.

One possible strategy used by researchers for recognition purposes relies on neural network (e.g., to recognize hand poses (Glauser et al., 2019) or microgestures (Sharma et al., 2021, 2023)). However, neural networks need a substantial amount of data to be trained. In our case, we would need to record several instances of data for each microgesture performed in each grasp context since the grasp will influence how the microgestures are performed. Thus, we opted for a simpler recognition strategy only using sensors placed at strategic positions. This type of recognizer has the added advantage of not having to be modified when a new grasp is considered.

Drawing inspiration from the DIY community, we propose a low cost and easy to build glove. Our glove prototype is covered with custom pressure-based sensors to recognize microgestures: one per phalanx on the front of the hand and one per joint of each finger on the back of the hand (Fig. 13). Sensors are made of Velostat™, a pressure-sensitive material which electrical resistance changes when being pressed. The three layers of Velostat™ are placed between two electrodes made of conductive fabric, inspired by existing instructables.⁶ Each sensor only requires basic sewing techniques and costs less than 1\$. Since the sensors are made of fabric and thin material, they offer little to no physical resistance while closing the hand. Fig. 13 represents the exploded view of a sensor. Each sensor is supplied with 5 V input, and the theoretical analog pin output values range is [0, 1024]. In practice, the range of values varies from one sensor to another, averaging around [300, 990]. As with industrial sensors, the lower limit of this range depends on how tightly the sensor is sewn onto the glove. The tighter it is sewn, the higher the pressure on the sensor, and therefore the higher the lower limit of this range.

The glove was designed for the left hand as this context is likely to happen when using microgestures in parallel with a primary task, e.g. piloting as in M[eye]cro (Wambecke et al., 2021) or rock climbing as exposed in the scenario of PinchWatch (Loclair et al., 2010).

7.2. Measuring glove impact and data collection

In this section, we want to make sure that wearing our DIY glove has little impact on the feasibility of microgestures. To do so, we performed a user study, reusing the design of our ergonomics factor study (study of Section 4), to compare the results with and without the glove. Moreover, we want to record enough data from the glove to design a microgesture recognizer.

⁶ <https://www.instructables.com/Stickytape-Sensors/>

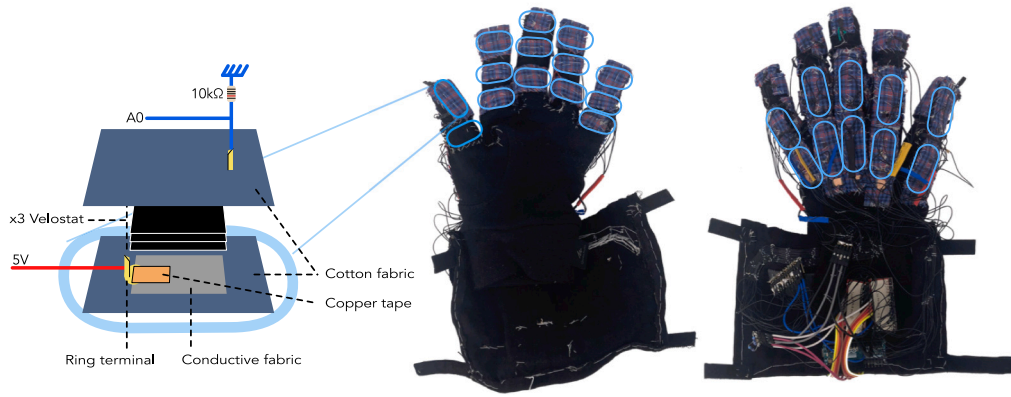


Fig. 13. Exploded view of a sensor and sensor positions on our glove. Sensors on the front of the hand are used to detect phalanx pressure. Sensors on the back of the hand are used to detect finger flexion. For each sensor, two electrodes made of conductive fabric are sewn on cotton fabric. A ring terminal is sewed on the conductive fabric to solder the wires from the Arduino. The copper tape increases the conductivity between the conductive fabric and the ring terminal. Between the electrodes, there are 3 layers of Velostat™. One electrode is connected to the 5 V pin of the Arduino, the other one to a voltage divider. The voltage divider is then connected to an analog input on the Arduino (e.g. the A0 pin).

7.2.1. Design

We designed a 13×21 within-subjects experiment with the independent variables *CONTEXT* and *MICROGESTURE*. We randomized the order of *CONTEXT*. For each *CONTEXT*, we randomized the order of *MICROGESTURE*. Each participant tested each pair *CONTEXT* $\times$ *MICROGESTURE* three times to record enough data to develop and test the recognizer.

The dependent variables are: if a pair *CONTEXT* $\times$ *MICROGESTURE* is *SKIPPED* or not and the *EASINESS* of the pair. We also recorded the *SENSOR DATA* from the glove while the microgesture was performed.

7.2.2. Measurements

Participants could skip a trial if the pair of *CONTEXT* $\times$ *MICROGESTURE* was not feasible. If the trial was not skipped, then we recorded the *EASINESS* of the pair using a 4-point Likert Scale (1 - hard, 4 - easy). We used an even number of graduations to avoid neutral answers.

7.2.3. Task

After presenting the purpose of the study, the participants sat down in front of a computer. For each *CONTEXT*, the experimenter explained to the participants how to hold the object. The experimenter then told the participants that they could perform the microgestures as they wished, but that they could not modify their grasp. For each pair *CONTEXT* $\times$ *MICROGESTURE*, a picture depicting the *CONTEXT* and a picture depicting the *MICROGESTURE* were displayed on the screen. Participants could then try to perform the microgesture. If the participant did not skip the pair, they then recorded three repetitions of the microgesture. Each recording is defined by a 3-second time window during which all raw sensor events are recorded. For each recording, the time window was started manually by participants by pressing a key on the keyboard. A progress bar displayed the time remaining. After the three repetitions, participants entered the easiness of the microgesture. After skipping the microgesture or recording the microgesture and entering its easiness, the next pair *CONTEXT* $\times$ *MICROGESTURE* was automatically displayed.

In all, the experiment lasted around 50 min.

7.2.4. Participants

We recruited 10 participants (Mean age = 25.3 yo, $\sigma = 5.12$), all using their right hand for daily interaction. Four participants were biologically female and six were male.

7.2.5. Results

The data recorded by the glove and the Likert scores are available in supplementary material. Fig. 14 presents the best median Likert score (scale from 1-hard to 4-easy) for each context $\times$ microgesture combination. In Fig. 14, crosses represent impossible combinations. If

a participant skipped a combination, no Likert score was recorded. If 50% or more participants skipped a combination, then the combination was deemed impossible. Median scores are therefore only calculated for combinations performed by at least 50% of the participants.

Compared with the results without the glove, we have differences in the feasibility of the microgestures. Given the glove bulkiness, thumb mob presses were feasible for pad grasps (both *Palmar* and *Tip Big*) because the thickness of the sensors made the thumb mob phalanx to touch the object. Moreover, for *Tip Small*, both *tap* and *press* of the thumb were feasible because the glove thickness made the sensors on the thumb and other fingers easier to reach. Similarly, for *Lateral Big*, *thumb mob tap* were feasible since it was easier to reach the object.

For *tap* and *press* only, the average Likert score without the glove is (3.8 ± 0.1 , $n = 72$) and with the glove is (3.5 ± 0.1 , $n = 76$). The confidence interval analysis suggests an effect of the glove on microgesture easiness. However, the cliffs δ suggests that this effect is medium ($\delta = 0.4$).

8. Recognizing a transferable actuator-based set of microgestures

8.1. Recognition factors

We want to recognize the actuator-based set, composed of seven microgestures: thumb tip, index tip, middle tip, all finger tips, thumb middle or base (mob), index mob and middle mob. Before creating the recognition system for this set, we examined the raw data collected from the gloves in the previous experiment. We identified three cases of ambiguity where we were unable to determine which finger had performed the microgesture, see Fig. 15.

First, the thumb cannot be used in the same set as the other fingers. In contexts where the thumb is opposite to the other fingers, i.e. pad grasps, a *press* of the thumb on the object will result in an increase in pressure on the other fingers, see Fig. 15A. This is similar to the freehand tap problem, where it is impossible to know the actuator when performing a *tap* of the thumb with a finger, because both fingers move and tap each other. We therefore decided not to consider the thumb as a possible actuator.

Second, *middle mob tap* and *middle mob press* in *Tip Small*, *Palmar Small*, and both *Lateral* contexts cannot be detected properly. In fact, to perform those microgestures, users tapped/pressed the middle finger on the index finger. Since the index finger was holding the object, this resulted in both a *middle mob tap/press* and an *index tip press*, see Fig. 15B. Therefore we removed *middle mob tap* and *middle mob press* for these four contexts.

Finally, for large objects, pressing one finger on the surface results in an increase of pressure of the other fingers to counterbalance, see

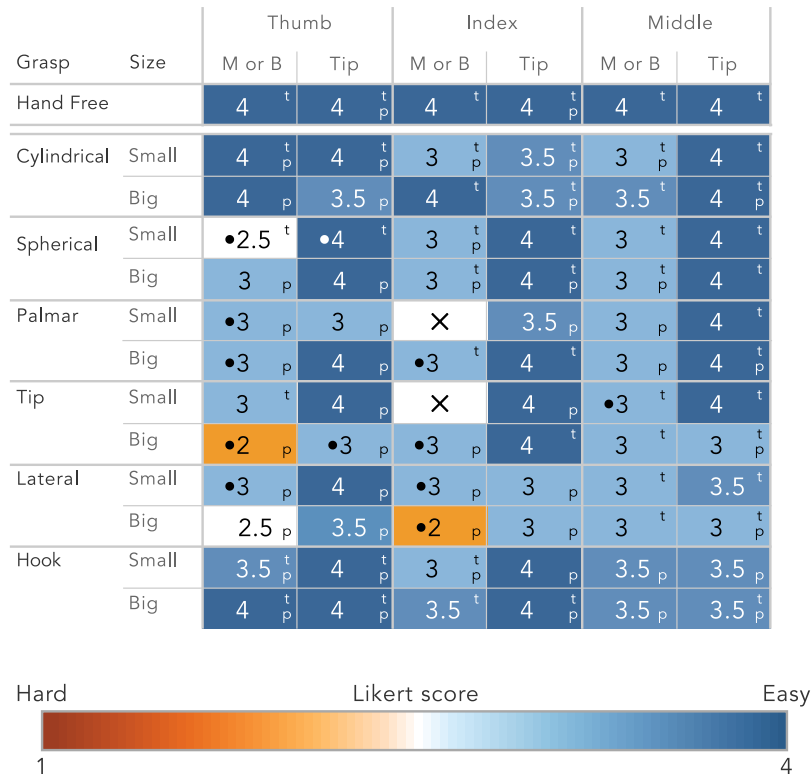


Fig. 14. For each context $\times$ microgesture combination: Median Likert score (scale from 1-hard to 4-easy). Cells with a $\times$ symbol correspond to context $\times$ microgesture combinations that were skipped by 5 or more participants. Cells with a $\bullet$ symbol correspond to combinations that were skipped at least once.

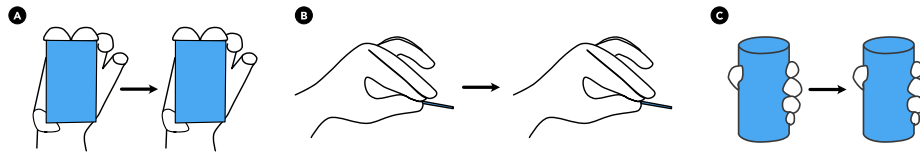


Fig. 15. (A) A thumb microgesture on the Palmar Big object also produces an index tip press and a middle tip press. (B) A middle mob microgesture on the back of the index in Tip Small also produces an index tip press. (C) A press with one finger (e.g. index finger as illustrated) for big objects also produces a press of all other fingers.

Fig. 15C. So, to avoid any ambiguity, we will guide the user to *tap* rather than *press* in these cases.

Our final actuator-based set is composed of *index mob*, *index tip*, *middle tip*, *middle mob* and *all tip*.

8.2. Microgesture recognizer

Our custom data glove is feeding raw values from each sensor to a recognizer system consisting of two software layers.

The first layer detects a microgesture for each sensor, without detecting if it is a *tap* or a *press*. Fig. 16, on top, shows the curves to be detected for a *tap* and a *press*. The two curves have similar shapes, but the *press* curve is shifted up. Since we do not need to know whether the user preforms a *tap* or *press*, we base our recognizer on the derivative of these curves, thus removing any upward shift.

The derivative of the two curves is shown at the bottom of Fig. 16. The derivative has two peaks, each corresponding to the rise and fall of the curve. We use two thresholds to detect the height of the peaks. These thresholds are the same for each user, but differ from sensor to sensor. They were set empirically using the data recorded in the previous experiment. Moreover, we added a duration constraint: the two peaks must occur within a time interval of 1 s. Finally, we recognize the finger and phalanx based on which sensor detected a microgesture.

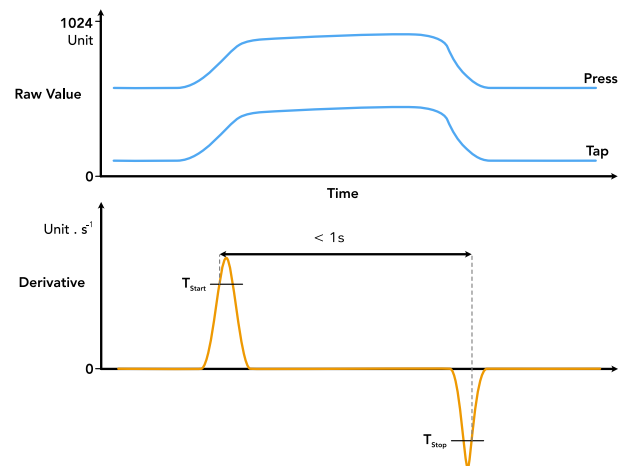


Fig. 16. Detecting a *tap* or a *press*. At the top, the two curves represent the raw values when performing a *tap* or a *press*. At the bottom, the curve represents the derivative of the raw values. A *tap* or a *press* is detected when the derivative goes above the threshold T_{Start} and then below the threshold T_{Stop} , within one second.

The second layer identifies the detected microgestures in a specific time window and decides which microgestures is recognized. When a

microgesture is detected, a time window of 200 ms starts. During this time window:

- if only one microgesture is detected, the system recognizes it;
- if a microgesture on at least three different fingers is recognized, an *all tips* microgesture is recognized;
- if a microgesture with the mob phalanx and a microgesture with the tip phalanx for the same finger are detected, the system recognizes only the microgesture with the tip phalanx.

Regarding the last heuristic, on large objects, a microgesture with the tip phalanx often results in a microgesture with the mob phalanx due to the mechanics of the hand. However, by keeping the finger straight, a microgesture with the mob phalanx can be performed without triggering a microgesture with the tip phalanx. Thus, when both a microgesture with the mob phalanx and a microgesture with the tip phalanx are detected, it is more likely that the microgesture with the tip phalanx is the correct microgesture.

After performing a microgesture, the hand will return to its initial position. This might trigger false positive. Thus, when a microgesture is recognized, there is a 1 s delay during which microgestures can no longer be recognized.

9. Recognition rates of transferable microgestures

We present a study that measured the recognition rate of the actuator-based set with our DIY glove and our recognizer.

9.1. Design

We designed a 13×5 within-subjects experiment with the independent variables *CONTEXT* and *MICROGESTURE*. For *CONTEXT*, we reused the same contexts as previously. For *MICROGESTURE*, we used the actuator-based set as defined in the previous section. We excluded 6 context $\times$ microgesture combinations: 2 because both *tap* and *press* are impossible, as shown in Fig. 12, and 4 due to recognition factors, as explained in Section 8.1. All the studied context $\times$ microgesture combinations are summarized in Figs. 17 and 18, cells with a cross correspond to the excluded ones.

We randomized the order of *CONTEXT*. For each *CONTEXT*, we randomized the order of *MICROGESTURE*. Each participant had to perform each *CONTEXT* $\times$ *MICROGESTURE* pair correctly three times.

The dependent variables are the *EASINESS* of the *CONTEXT* $\times$ *MICROGESTURE* pair and if the microgesture was recognized with *SUCCESS*.

9.2. Measurements

After correctly performing the microgesture three times, the participant entered the *EASINESS* of the microgesture using a 4-point Likert Scale (1 - hard, 4 - easy). We used an even number of graduations to avoid neutral answers. The *SUCCESS* was computed based on the recognizer's output.

9.3. Task

After presenting the purpose of the study, the participants sat down in front of a computer. For each *CONTEXT*, the experimenter explained to the participants how to hold the object. Next, the experimenter guided participants on how to perform a microgesture using the following explanations:

- prefer to use a *tap* rather than a *press*;
- when pressing with the index or middle finger, be aware of where and what you are doing with the other fingers;
- keep your finger straight when performing a microgesture with the mob phalanx. It will prevent tip microgestures;

- with small objects, go around the object to perform a microgesture on your thumb;
- for *Lateral* contexts, use your thumb to perform a *press* through the object on the desired index part.

We have allowed small variations in the grasp, provided that the grasp remains unchanged according to Schlesinger's taxonomy.

For each pair *CONTEXT* $\times$ *MICROGESTURE*, a picture depicting the *CONTEXT* and a picture depicting the *MICROGESTURE* were displayed. The participants had a short *training* phase for each pair *CONTEXT* $\times$ *MICROGESTURE*. During this *training* phase, the participants were guided on which microgesture to perform and where (a phase that generally lasted less than a minute). After two correct recognitions of a microgesture, participants started the *experiment* phase. During this phase, participants had to perform the microgesture correctly three times. If they failed to perform a correct microgesture, the trial was flagged as an error and participants had to repeat it immediately. To start a trial, participants pressed the keyboard S key. For each repetition, participants had 3s to perform a microgesture. A visual feedback (screen background turning green or red) indicated whether the recognized microgesture was correct or not.

In all, the experiment lasted 45 min.

9.4. Participants

We recruited 10 participants (Mean age = 26.1 yo, $\sigma = 2.92$) who perform daily interaction tasks with their right hand. Four participants were biologically female and six male. Prior to the experiment, we insured that the glove was suitable for the participants. Users with small hands were excluded if the glove slipped on their hand and the sensors on the phalanxes were no longer in place. Users with large hands were also excluded if the glove restricted their movements too much. $10 \text{ (participants)} \times 65\text{-}6 \text{ (context} \times \text{microgesture combinations)} \times 3 \text{ (repetitions)} = 1770$ error-free data points were collected. We also collected 252 erroneous data points.

9.5. Results

The data analysis follows recommendations for "fair statistical communication in HCI" (Dragicevic, 2016). Data were aggregated per user then per pair of context $\times$ microgesture. Reported confidence intervals are 95% confidence intervals computed with the formula $CI = 1.96 * \frac{\sigma}{\sqrt{n}}$, with σ the standard deviation and n the size of the considered sample. We also report the proportion overlap (POL) (Cumming, 2009) when comparing two CIs, i.e. overlap amount/mean size of the CI arms that overlap. Using the TIP 24 of the guidelines (Dragicevic, 2016), if $POL < 0.25$, then a reasoning similar to that of a p -value less than 0.05 can be used.

Given that our experimental protocol required that a given microgesture be performed within a 3s time window, some trials ended without any microgesture being recognized. Since we cannot differentiate trials in which participants were not ready and those in which participants attempted a microgesture but no microgesture was detected, we removed these trials (29 trials from 1770 trials) for further analysis. However before doing so, we first validated that these trials play a marginal role in the results. Including these trials, the overall microgesture recognition rate is 93.3%, with CI of $\pm 1.3\%$, computed for 59 microgestures. Excluding these trials, the overall microgesture recognition rate is 94.3%, with CI of $\pm 1.4\%$, computed for 59 microgestures. The proportion overlap (POL) between both CIs is 1.3. Thus, there is no apparent difference between taking these trials into account or not. In the following, the mean recognition rates (or the average of median Likert scores), their respective CI and the number of microgestures used to compute the mean, will be formatted as follows ($94.3\% \pm 1.4, n = 59$).

Fig. 17 presents the mean recognition rates for all context $\times$ microgesture combinations.

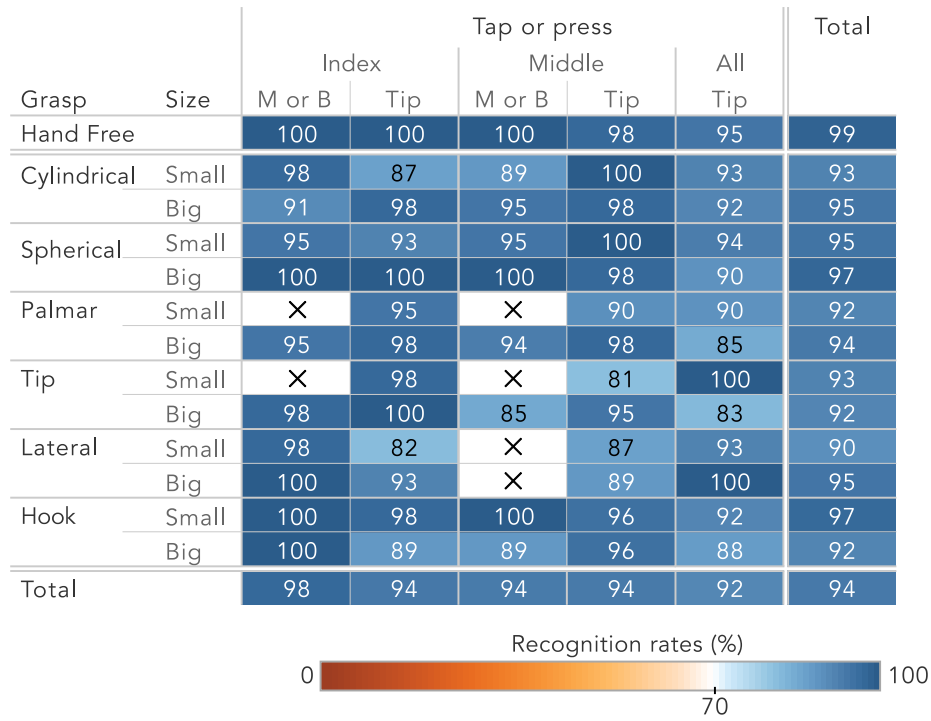


Fig. 17. Mean recognition rates for each microgesture in a specific context. X represents an impossible combination.

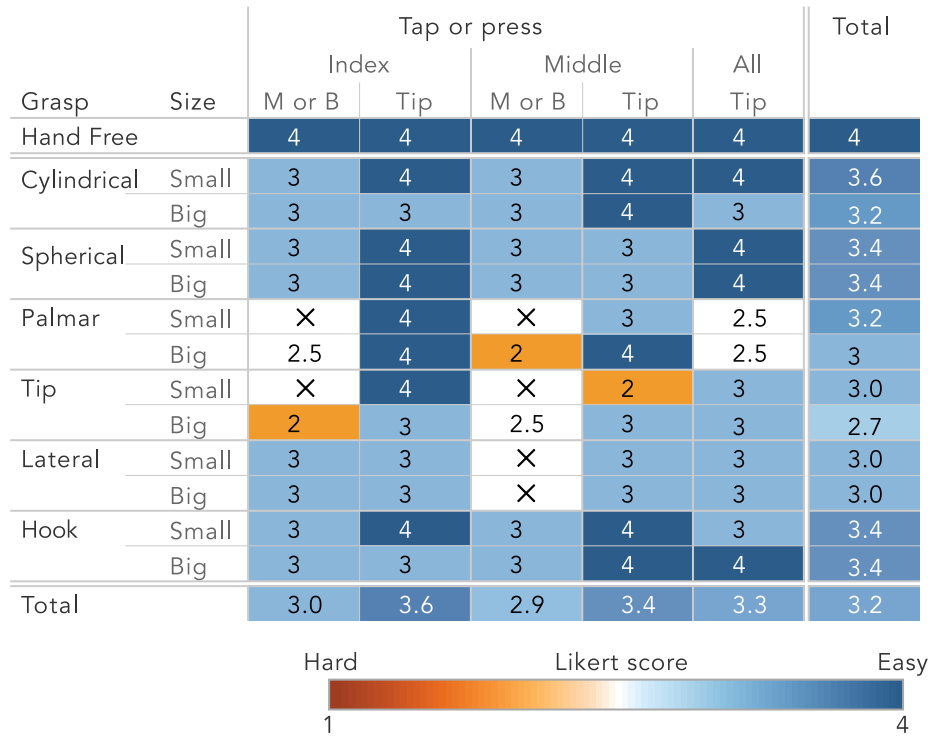


Fig. 18. Median Likert scores for each microgesture in a specific context. X represents an impossible combination.

Fig. 18 presents the median easiness Likert scores (scale from 1-hard to 4-easy) for all context $\times$ microgesture combinations. In total, the average median Likert score across the 13 contexts is (3.2 ± 0.2 , $n = 59$).

The actuator-based set has a high recognition rate, ($94.3\% \pm 1.4$, $n = 59$), with the worst context being *Lateral Small* ($90\% \pm 1.4$, $n = 4$), the best contexts being *Spherical Big* ($97\% \pm 4.0$, $n = 5$) and *Hook Small* ($97\% \pm 2.9$, $n = 5$), which have a similar recognition rate as the *Hand Free* ($99\% \pm 2.0$, $n = 5$),

both POL being under -1.4 . Thus, our low-cost DIY glove is a usable solution for user experiments and allows us to perform microgestures on different types of surfaces, *i.e.* hand surface or held object surface. However, the glove can be simplified. We have 24 sensors to detect pressure on every phalanx and the flexion of two phalanx per finger. To detect the actuator-based microgesture set, it is sufficient to place the sensors on each phalanx of the index and middle fingers and one sensor on the tips of the other fingers. Overall, considering the actuator

instead of the action expands the set of transferable microgestures by considering the combination of actuators and, in our case, is highly recognizable with a simple data glove and a simple recognizer.

10. Overall discussion and future work

In this section, we discuss the results of our ergonomics factors study (Section 4) and our recognition study (Section 9).

10.1. Distinguishing a phalanx in a grasp context

We introduced the use of the middle or base (mob) phalanx of a finger in a grasp context for *tap* and *press*. To our knowledge, previous works only studied different phalanxes in a free-hand context (Soliman et al., 2018; Whitmire et al., 2017; Way, 2014; Deyle et al., 2007; Wambecke et al., 2021). In grasp context, the closest studied property is the side of the finger (Joshi et al., 2023).

On the feasibility side, the mob phalanx is usable as much as the tip phalanx for *tap* but not for *press*. For *press*, with grasps where the object is held between the fingertip, only the *tip press* is available. On the easiness side, there is a plausible difference between the tip phalanx (4 ± 0 , $n = 13$) and the mob phalanx (3.5 ± 0.3 , $n = 13$). That effect is however small (Cliff's $\delta = 0.32$). While this difference is highly plausible for *Cylindrical Small* $\times$ *middle finger*, POL of -4.8 , *Palmar Big* $\times$ *middle*, POL of -1.9 , and might be plausible for *Palmar Big* $\times$ *index*, POL of 0.3 , the difference is unlikely for any other pairs of context $\times$ microgesture (all POL > 0.5). On the recognition side, there is no plausible difference between the tip phalanx (94.5 ± 2.2 , $n = 26$) and the mob phalanx (96.1 ± 2.0 , $n = 20$), POL of 1.2 .

To summarize, considering different phalanxes, as in a free-hand context, is possible in grasp contexts.

10.2. Using a virtual finger analysis rather than grasps

As in previous works (Sharma et al., 2019, 2021, 2023; Saponas et al., 2009), we studied microgestures in grasp contexts reusing the taxonomy of Schlesinger. However, by also considering virtual fingers, we came to the conclusion that grasps with the same virtual finger analysis performs similarly. For instance, *Palmar Small* and *Tip Small*, or *Palmar Big* and *Tip Big*, have the exact same impossible microgestures, similar easiness scores and similar recognition rates. The only exception is the *Spherical Big* context which is slightly harder than both *Cylindrical* contexts. However, we believe that this is due to the size of the *Spherical Big* object. Since it was too big for some participants, the virtual fingers for these participants were actually



instead of the theoretical virtual fingers:



In the end, for a same virtual finger arrangement, results are similar, whereas for a given grasp context results can vary according to object size, i.e. *Palmar Big* vs *Palmar Small*, or the hand sizes, i.e. *Spherical Big* for smaller vs bigger hands. Therefore, instead of testing microgestures for each grasp of Schlesinger's taxonomy or GRASP taxonomy, a better

approach might be to study different virtual finger arrangements. This is all the more important when studying a specific object, since the same object can be held with different virtual finger arrangements.

10.3. Human capabilities: exploring grasp switching

In our study, which focused on transferable microgestures, participants were not allowed to modify their grasp. However, future work could consider changing the grasp to increase the number of feasible and easy-to-perform microgestures. For instance, a pen can be held as a *Tip Big* or can be easily switched to a *Cylindrical Small*. However, while in the *Cylindrical Small* contexts all microgestures are feasible with ease (4 ± 0 , $n = 21$), in *Tip Big*, 6/21 microgestures are impossible and the remaining microgestures are difficult to perform (3.6 ± 0.2 , $n = 15$), POL = -0.1 . As a result, with a pen, a user may be tempted to change grasps for easy access to more commands. However, the question of when and why a user changes grasps when performing microgestures remains open. A first answer can be found in the work of Song et al. (2011) who considered grasps as inputs in combination with microgestures. In this case, users change the way they hold the pen to access different modes.

10.4. Transferable microgestures: action-based or actuator-based?

We have described two approaches to creating a set of microgestures that are transferable across grasp contexts. While previous work has focused on the action of microgesture, which can be performed by any actuator (Sharma et al., 2019; Joshi et al., 2023) (action-based), our approach is based on the actuator (i.e. the finger): microgestures are distinguished by the actuators, not by the actions performed by these actuators (actuator-based). Both approaches can be used. If we consider a small set of commands such as: play/pause and next/previous. With the action-based approach, play/pause can be associated to a *tap*, and next/previous can be associated to a *swipe forward/inward*. With the actuator-based approach, play/pause can be associated to an *all tip*, and next/previous to an *index/middle tip*. Also mentioned in Chaffangeon Caillet et al. (2023), future studies are required to identify which approach is the best in terms of memorability and usability.

Nonetheless, both approaches have their limits. Sets from the free-hand microgesture literature can comprise more than 15 microgestures (Chan et al., 2016; Soliman et al., 2018; Lee et al., 2020), whereas in the literature on grasping microgestures, the largest set of transferable microgestures comprises only 9 microgestures (Sharma et al., 2019). It is possible to merge the two approaches by considering different combined actions, such as *tap/press*, and different actuators. There are still unexplored ways to increase the size of transferable microgesture sets.

11. Conclusion

In this paper, we presented knowledge and methods for identifying and recognizing microgestures that can transfer across contexts.

Our first study considers the most frequently studied microgestures, i.e. *tap*, *press*, *stretch* and *vertical swipe*, and distinguishes them according to their actuator, as has been done previously and separately in the microgesture literature on grasping and on the free-hand. For the first time, *tap* and *press* were distinguished based on the actuator surface that touches/presses during the microgesture, not only for the free-hand context, but also for the grasp contexts. All six grasps of Schlesinger's taxonomy, commonly used for interaction studies, were studied with two sizes of objects. We also considered the hand-free context resulting in 13 contexts for our studies. We identify and report on impossible combinations of context $\times$ microgesture, as well as the ease of execution of feasible combinations.

To avoid future feasibility studies that require times and effort to do, we propose a fast and easy way of determining the feasibility of a microgesture by relying on the microgesture notation μ Glyph (Chaffangeon

Caillet et al., 2023). With this conceptual method, we were able to correctly determine the feasibility of microgestures in the first experiment 99% of the time.

The set of microgestures we obtained from the experimental study and from the conceptual analysis were not transferable across all our contexts, *i.e.* some microgestures were not feasible in some contexts. After presenting two ways of defining a set of microgestures transferable across our contexts, we presented an actuator-based set of microgestures where *tap* and *press* are combined into a single microgesture. This set is composed of 4 single finger microgestures (index finger tip, index finger middle or base, middle finger tip and middle finger middle or base) and a multi-finger microgesture, *i.e.* all tips.

To recognize this actuator-based set of microgestures, we built a low-cost DIY glove. We first checked the impact of the glove on the microgesture feasibility and easiness. Due to the glove bulkiness, microgestures were overall harder to perform and 6 context $\times$ microgesture became feasible. With this glove, we obtain a microgesture recognition rate of 94% for the 13 free-hand and grasp contexts and without augmenting the held objects. Therefore, this glove allows us to perform microgestures on any surface, *i.e.* hand surface or held object surface.

CRediT authorship contribution statement

Adrien Chaffangeon Caillet: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Alix Goguy:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Laurence Nigay:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ijhcs.2024.103398>.

Data availability

Data will be made available on request.

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